

Evaluation in Information Retrieval

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Motivation

- ▶ IR is an *empirical* discipline.
- ▶ Intuition can be wrong!
 - ▶ “Sophisticated” techniques need not be the best
e.g., rule-based stemming vs. statistical stemming
- ▶ Proposed techniques need to be validated and compared to existing techniques.

Cranfield Method (Cleverdon et al., 1960s)

Benchmark data

- ▶ Document collection
- ▶ Query / topic collection
- ▶ Relevance judgments – information about which document is relevant to which query

Assumptions

- ▶ Relevance of a document to a query is objectively discernible
- ▶ All relevant documents in the collection are known
- ▶ All relevant documents contribute equally to the performance measures
- ▶ Relevance of a document is independent of the relevance of other documents

Evaluation Metrics

Background

- ▶ User has an information need.
- ▶ Information need is converted into a **query**.
- ▶ Documents are **relevant** or **non-relevant**.
- ▶ Ideal system retrieves all and only the relevant documents.

Set-based Metrics

$$\mathbf{Recall} = \frac{\#(\text{relevant retrieved})}{\#(\text{relevant})} = \frac{\#(\text{true positives})}{\#(\text{true positives} + \text{false negatives})}$$

$$\mathbf{Precision} = \frac{\#(\text{relevant retrieved})}{\#(\text{retrieved})} = \frac{\#(\text{true positives})}{\#(\text{true positives} + \text{false positives})}$$

Metrics for Ranked Results

(Non-interpolated) average precision

Which is better?

1. Non-relevant
2. Non-relevant
3. Non-relevant
4. Relevant
5. Relevant

1. Relevant
2. Relevant
3. Non-relevant
4. Non-relevant
5. Non-relevant

Metrics for Ranked Results

(Non-interpolated) average precision

Rank	Type	Recall	Precision
1	Relevant	0.2	1.00
2	Non-relevant		
3	Relevant	0.4	0.67
4	Non-relevant		
5	Non-relevant		
6	Relevant	0.6	0.50
∞	Relevant	0.8	0.00
∞	Relevant	1.0	0.00

$$AvgP = \frac{1}{5} \left(1 + \frac{2}{3} + \frac{3}{6} \right)$$

(5 relevant docs. in all)

$$AvgP = \frac{1}{N_{Rel}} \sum_{d_i \in Rel} \frac{i}{Rank(d_i)}$$

Metrics for Sub-document Retrieval

- ▶ Let p_r be the document part retrieved at rank r
- ▶ $rsize(p_r)$: amount of relevant text contained in p_r
- ▶ $size(p_r)$: total number of characters contained in p_r
- ▶ T_{rel} : total amount of relevant text for a given topic

$$P[r] = \frac{\sum_{i=1}^r rsize(p_i)}{\sum_{i=1}^r size(p_i)}$$

$$R[r] = \frac{1}{T_{rel}} \sum_{i=1}^r rsize(p_i)$$

Metrics for Ranked Results

- ▶ **Precision at k ($P@k$)** – precision after k documents have been retrieved
 - ▶ easy to interpret
 - ▶ not very stable / discriminatory
 - ▶ does not average well
- ▶ **R precision** – precision after N_{Rel} documents have been retrieved

Cumulated Gain

Idea:

- ▶ Highly relevant documents are more valuable than marginally relevant documents
- ▶ Documents ranked low are less valuable

$$Gain \in \{0, 1, 2, 3\}$$

$$G = \langle 3, 2, 3, 0, 0, 1, 2, 2, 3, 0, \dots \rangle$$

$$CG[i] = \sum_{j=1}^i G[j]$$

Normalised Discounted Cumulative Gain (nDCG)

$$DCG_p = \sum_{i=1}^p \frac{G[i]}{\log_2(i+1)}$$

Ideal $G = \langle 3, 3, \dots, 3, 2, \dots, 2, 1, \dots, 1, 0, \dots \rangle$

$$nDCG[i] = \frac{DCG[i]}{\text{Ideal } DCG[i]}$$